# Datamines India Internship

Department of Computer Science and Engineering

# STUDENT PLACEMENT PREDICTION SYSTEM USING MACHINE LEARNING

**A PROJECT REPORT**

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## Abstract

*In the competitive landscape of higher education, securing campus placements is a primary goal for students and institutions. This project develops a "Student Placement Prediction System" using Machine Learning to predict the likelihood of a student being placed based on academic and technical parameters. Utilizing a Random Forest Classifier, the system processes historical data including CGPA, internship experience, and technical scores. We introduce custom feature engineering metrics such as Average Score and Skill Aggregation to enhance predictive accuracy. The final model is deployed via a Flask-based web application, providing students with both placement predictions and actionable recommendations to improve their profiles.*


## 1. Introduction

Campus recruitment is a pivotal event in a student's academic journey. However, many students remain unaware of how their current academic standing and skill sets align with industry requirements. Traditional methods of evaluation often overlook the synergy between academics and practical skills.

This project aims to bridge this gap by leveraging data-driven insights. By analyzing historical placement data, the system identifies patterns that lead to successful recruitment. The core of this system is a Machine Learning model that classifies students into "Placed" or "Not Placed" categories, acting as an early warning and guidance tool.

## 2. Problem Statement

Students often lack a quantitative understanding of their placement readiness. Academic institutions also struggle to identify "at-risk" students who may require additional training. There is a need for an automated system that can:

- Analyze multifaceted student data (CGPA, Skills, Projects).
- Predict placement outcomes with high accuracy.

- Provide insights into which specific areas a student needs to improve.

## 3. Objectives

- To collect and preprocess student performance data from reliable sources like Kaggle.
- To implement custom feature engineering to better represent student capability.
- To build a robust classification model using the Random Forest algorithm.
- To develop a user-friendly web interface using Flask for real-time predictions.
- To provide a recommendation logic that guides students based on their weaknesses.

## 4. Literature Review

Recent studies in Educational Data Mining (EDM) show that algorithms like Decision Trees, SVM, and Random Forest are effective in predicting student performance. Previous researchers focused primarily on CGPA; however, our study expands this by incorporating internships, project counts, and soft skills (communication, aptitude), which are increasingly valued by modern recruiters. Random Forest was chosen for this project due to its ability to handle non-linear relationships and its resistance to overfitting.

## 5. System Architecture

The system follows a modular architecture consisting of the Data Layer, Logic Layer (Model), and Presentation Layer (Web App).

```
+----------------------------------------+       +----------------------------------------+
|       USER INTERFACE      | <--> |       FLASK BACKEND       |
|    (HTML/CSS Dashboard)   |       |    (Python Controller)    |
+----------------------------------------+       +-------------+--------------------+
                                                                |
                                                                v
+----------------------------------------+       +-------------+--------------------+
|    RECOMMENDATION LOGIC | <--> |    ML MODEL (RF .pkl)     |
|    (If-Else Heuristics)   |       |    (Predictor Engine)     |
+--------------------------+       +-------------+--------------------+
                                                                |
                                                                v
                                          +-------------+--------------------+
                                          |    PREPROCESSED DATA     |
                                          |  (Feature Engineering)   |
                                          +----------------------------------------+
```

## 6. Workflow of the System

The step-by-step process of the system is illustrated below:

```
                                  [ START ]
                                       |
                                       v
        [ Input: CGPA, Internships, Projects, Aptitude, Coding, Comm. ]
                                       |
                                       v
      [ Feature Engineering: Calculate avg_score, total_skills, is_weak ]
                                       |
                                       v
```

```
[ Load Trained Random Forest Model (.pkl) ]
                |
                v
[ Generate Prediction: Placed / Not Placed ]
                |
                v
[ Apply Recommendation Logic ]
                |
                v
[ Display Result on Dashboard ]
                |
                v
[ END ]
```

# 7. Dataset and Data Preprocessing

## 7.1 Dataset Description

The system utilizes a student placement dataset sourced from **Kaggle**. It contains features such as CGPA, Number of Internships, Projects, Aptitude Test Score, Coding Score, and Communication Skills.

## 7.2 Data Preprocessing Steps

- **Data Cleaning:** Handling missing values through mean imputation.
- **Encoding:** Converting categorical data (e.g., Stream, Gender) into numerical values using Label Encoding.
- **Scaling:** Using Standard Scaler to normalize numerical features for better model convergence.

## 8. Feature Engineering

To improve the model's decision-making, three custom features were engineered:

1. **Average Score (avg_score):** Calculated as `(CGPA + Internships + Projects) / 3`. This represents a holistic score combining academic and practical exposure.
2. **Total Skills (total_skills):** Calculated as `Aptitude + Coding + Communication`. This aggregates the student's technical and soft skill readiness.
3. **Weak Candidate Flag (is_weak):** A boolean flag where `is_weak = 1` if `CGPA < 6`, else `0`. This helps the model penalize low academic performance regardless of other skills.

## 9. Model Building and Evaluation

### 9.1 Algorithm: Random Forest Classifier

Random Forest is an ensemble learning method that constructs multiple decision trees during training. It outputs the mode of the classes for classification. It was chosen for its high accuracy and ability to rank feature importance.

### 9.2 Train-Test Split

The dataset was split into **80% Training** and **20% Testing** sets to ensure the model generalizes well to unseen data.

### 9.3 Evaluation Metrics

- **Accuracy:** The model achieved an impressive accuracy of ~88-92% on the test set.
- **Classification Report:** Precision, Recall, and F1-Score were analyzed to ensure the model does not unfairly bias against "Not Placed" students.

## 10. Implementation Details

- **Backend:** Flask (Python framework) handles the routing and model inference.

- **Frontend:** Responsive UI built with HTML5 and CSS3.
- **Model Storage:** The trained Random Forest model is serialized using the `pickle` library as `model.pkl` for fast loading.

## 11. Recommendation System Logic

If the model predicts "Not Placed", the system triggers a logic engine:

- If **Internships < 1**: Suggest "Apply for summer internships to gain industry exposure."
- If **Coding Score < 60**: Suggest "Practice Data Structures and Algorithms on platforms like LeetCode."
- If **Communication < 5**: Suggest "Participate in mock interviews and group discussions."

## 12. Future Scope

Future enhancements include integrating LinkedIn API to fetch real-time student skills and using Deep Learning (ANN) for more complex pattern recognition.

## 13. Conclusion

The Student Placement Prediction System successfully demonstrates the application of Machine Learning in the educational sector. By using Random Forest and custom feature engineering, we provide a tool that not only predicts outcomes but also serves as a career counselor for students.

## 14. References


1. Kaggle Dataset: Student Placement Data (2023).
2. Scikit-Learn Documentation: Random Forest Classifier.
3. Flask Documentation: Web Development with Python.